

STEM Course Prep
In Class Activities
Chapter 7

Slides 2-9

1. Below is a list of student names. Create a matrix (vertically concatenate) composed of these names. Use two methods.

Kim, John, Sara, Steven

```
name_mat=['Kim ' ; 'John ' ; 'Sara ' ; 'Steven']  
name_mat2=char('Kim', 'John', 'Sara', 'Steven')
```

Now display the contents found in various elements of name_mat2.

```
name_mat2(4,2)  
name_mat2(1,5)  
name_mat2(3,3)
```

2. Horizontally concatenate the following words: Blue, Green. Use two methods.

```
string1= ['Blue' 'Green']  
string1=strcat('Blue', 'Green')
```

Slides 10-12

3. Plot the response for an underdamped second order system for a time range of 0-8 seconds. Find the overshoot (max value of response) of $c(t)$ and use that value to customize the plot title.

$$c(t) = 1 - e^{-t} \left(\cos(\sqrt{8}t) + \frac{\sqrt{8}}{8} \sin(\sqrt{8}t) \right)$$

```

t=0:.05:8;
c=1-exp(-t).*(cos(sqrt(8)*t)+(sqrt(8)/8)*sin(sqrt(8)*t));

%find data to be used in plot labels
mint=min(t);
maxt=max(t)
maxc=max(c);

plot(t,c,'mp')
axis tight
xlabel('Time (seconds)')
ylabel('Response (m)')
string=sprintf('Underdamped Response from %d seconds to %d ...
seconds\nOvershoot = %.3f',mint,maxt,maxc)
title(string)

```

Slides 16-17

4. Compare the following strings: 'Compressible Flow', 'Incompressible Flow', 'compressible Flow'

```

string1='Compressible Flow';
string2='Incompressible Flow';
string3='compressible Flow';
result=strncmp(string1,string2,3)
result=strncmpi(string1,string3,3)
result=strcmpi(string1,string3)

```

Slides 18-21

5. Ask the user to input the name of a required ENGR or AERO course. If the input is an ENGR course (ENGR 2301, 2302, 2332) change the prefix to AERO. If the input is an AERO class do nothing.

```

class=input('Enter the name of your Engineering class:','s');
if isempty(strfind(class,'ENGR'))
    fprintf('Your course is unchanged\n')
else
    new_class=strrep(class,'ENGR','AERO')
    fprintf('Your course has changed! New course name: %s\n',new_class)
end

```

6. Copy the three files listed: a0v100.txt, a10v100.txt, a20v100.txt. Use the ls function and a for loop to load the three data files into MATLAB.

```
files= ls ('*v100.txt')
for i=1:3
    load (files(i,:))
end
```

Note: the * tells MATLAB to look for any file ending in v100.txt

7. Copy time1.txt, time2.txt, time3.txt. These files contain three columns of data: time, Vx, Vy. These files were created to look at the velocity for a projectile. The data was broken up and put into three different files. Time1.txt contains data for time 0-4 sec, time2 contains data for 4.05 – 8 seconds, time3 contains data for 8.05 - 10 seconds. Use the ls function and a for loop to plot the data (time vs. Vy) from the three files.

```
time_files= ls('time*.txt')
[numFiles,cols]=size(files); % see how many files there are
data=[]; % initialize the matrix that will be used to store plot data

for i=1:numFiles
    temp=load(time_files(i,:));
    data=[data;temp]; %concatenate data
end
plot(data(:,1),data(:,3),'mp')
xlabel('Time (sec)')
ylabel('Vy (m/s)')
string=sprintf('%d Data Points from %d Files',length(data),i)
title(string)
axis tight
```

Slides 23-24

8. Create a plot of two user-selected plots. The user can select between four options: cos, sin, log, sqrt. The user also must input the color of each plot. Use the strfind and strcat commands to create the plot.

```
func=input('Enter the function to plot... (log,sqrt,sin,cos):\n','s');
func2=input('Enter the second function to plot... (log,sqrt,sin,cos):\n','s');
color1=input('Enter the color for plot 1 (r, m, c, y,... b):\n','s');
color2=input('Enter the color for plot 2 (r, m, c, y,... b):\n','s');
hold on
if isempty(strfind(func,'cos')) & isempty(strfind(func,'sin'))
    a=0:.01:10;
    plot(a,eval(strcat(func,'(a)')),color1,'linewidth',3)
    plot(a,eval(strcat(func2,'(a)')),color2)
else
    a=0:0.01:2*pi;
    plot(a,eval(strcat(func,'(a)')),color1,'linewidth',3)
    plot(a,eval(strcat(func2,'(a)')),color2)
end
legend(func,func2)
```